

Experimental Weaving at the Unstable Design Lab

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Figure 1: an overview of interactive textile demos and software for textile design to be showcased.

Abstract

This demo showcases "experimental weaving" as it has been explored by researchers and experimental weavers in residence at the Unstable Design Lab. The demo will feature interactive woven textiles, software to support complex woven structure design, and instructional resources for visitors to explore within their research.

Keywords

interactive textiles, weaving, computational design

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1 Introduction and Relevance

Over the past decade, an increasing amount of research that leverages textile techniques and equipment has been showcased at CHI: from bringing new forms of interactivity via soft and flexible cloth interfaces to understanding the inherent support for 3D and multi-material assemblies by textile equipment such as looms, and embroidery and knitting machines. The Unstable Design Lab, housed within the ATLAS Institute at CU Boulder, has been participating in the integration of textiles into CHI since 2016 with a focus on woven textiles. We aim for this demo to showcase what we have developed in that time. This includes interactive textile prototypes such as **Magnetic Reverberations** [12], **A Fabric that Remembers** [7], **Wear** [2], and several others [8, 11, 15]. Additionally, we will show **AdaCAD**, an open-source software tool we developed to aid in the development of complex 3D textile structures that has been

adopted by hundreds of researchers and artists [9, 10]. Alongside these tools and e-textile interfaces, we will showcase resources we have produced to support other researchers in integrating weaving in their research: from the **Experimental Weaving Residency**, a program for including craft weavers into our research lab [5, 6], to the **Introduction to Weave Structure for HCI workbook**, a how-to resources for HCI researchers to use to more deeply understand the potential of (woven) textiles [7]. This interactivity demo will bring attention to our research and findings in formats that are both engaging and instructive. At the same time, the focus on "experimental weaving" highlights our belief that the inclusion of artists and craftspeople in research exploring new innovations in interactive textiles brings additional strength to textiles-based research.

2 Broader Context and Questions

Researchers looking to integrate textiles into their research face challenges with material variability, vocabulary, developing and communicating embodied knowledge, and high knowledge barriers when advancing beyond basic techniques and/or maintaining weaving equipment [1, 13]. We address these challenge in our research by actively supporting and collaborating with "experimental weavers" through a program we developed called the Experimental Weaving Residency [5]. Established in 2019, the Experimental Weaving Residency is a program that hosts weavers in our research lab for up to three months with the goal of creating resources and techniques to share with a broader community. We use the term "experimental weaver" and "experimental weaving" to bring attention to complex 2D and 3D woven forms created by weavers who work closely with hand and/or automated machinery with an eye towards pushing the boundaries of woven structure and form. Experimental weavers are trained through art, craft and design programs and many have advanced degrees in textiles or fiber arts. These include prior residents of our program, like Elizabeth Meiklejohn who explores how partial weft insertions can give rise to textiles with bifurcated surfaces [12], or Kathryn Walters, a PhD student at the Swedish School of Textiles, who explores how textiles made with overtwisted linen change shape and texture when exposed to humidity [14]. These two exemplars are just a few of the people pushing the boundaries of woven structure and materiality who sit on the margins or outside of HCI. Our demo aims to bring attention to both the relevance of weaving for HCI and the relevant skill-sets of the experimental weaving community while also offering concrete examples for how researchers to engage.

We do this by showcasing the creative work that emerged through or in parallel to our Experimental Weaving Residency and by providing materials and links to resources that describe both the structure and outcomes of these programs that others can review, and perhaps replicate. Ultimately, our goal for this demo is to raise awareness of the potential of woven structure and electronics integration in textiles, to bring visibility to the resources we have produced to support research in this area and to advocate for the inclusion of artists and craftspeople ("experimental weavers") in materials-led innovation. The specific demos and resources we plan to showcase are listed in the sections below.

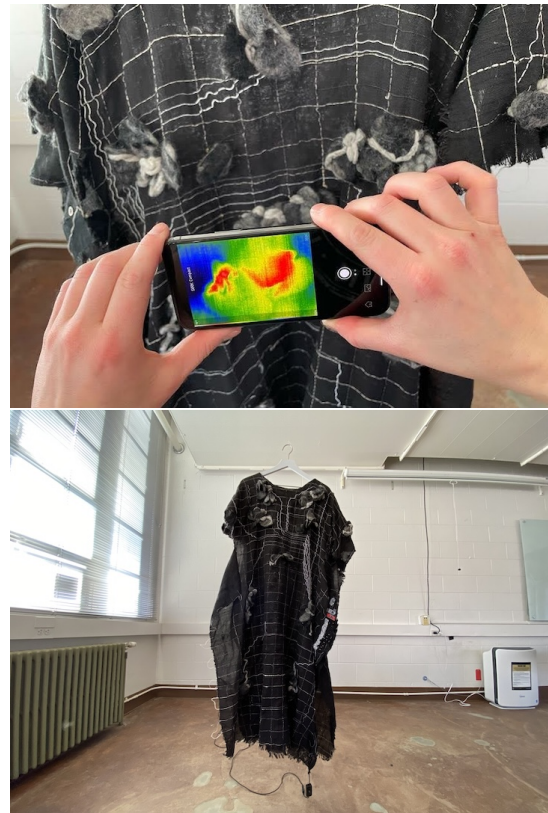


Figure 2: Wear: an interactive house dress that plays back interactions captured on the body through heat. Participants engage the demo by viewing the garment through an infrared camera.

3 Walkthrough

Visitors to our demo will see a vibrant space with interactive textiles, material samples, and software interfaces. On one side of the booth, we'll feature hands-on *interactive demos* showcasing the potentials for sensing and actuation in textiles. On the other, we'll focus on *resources* we have developed for the community to engage experimental weavers and/or weaving. This includes a live demo of the AdaCAD software as well as a collection of texts, textile samples, how-to workbooks, exhibition catalogs, and videos that have emerged from our prior work and residency program. Our goal is to both engage the community in the "bread and butter" of CHI (interactivity) and then equip interested visitors with a series of resources that can help them take next steps towards integrating these techniques in their practice.

3.1 Interactive Physical Prototypes

We will include demonstrations of three interactive textile prototypes that showcase different methods and outcomes of embedding circuitry in textile form. These will include:

3.1.1 Wear. Wear is a project that chronicles a year spent making, measuring, and marking experiences as a mother during COVID

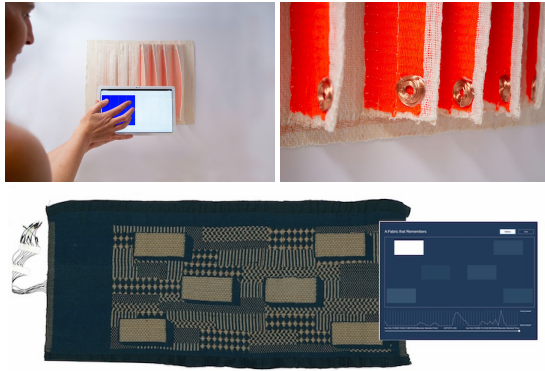


Figure 3: Left: Magnetic Reverberations: electromagnets integrated into the cloth allow a visitor to control the movement and coloring of the cloth via a touch controller. Right: A Fabric that Remembers: Visitors interact by pressing the cloth on its force sensing press regions. The presses are recorded in real-time on a digital interface that allows you to scroll back in time, revealing this history of touch on the cloth.

lockdown [3]. It is a Design Memoir [4] in the form of a handmade woven garment with integrated circuitry that measures the forces on first-author's body. It accompanied by a video that documents, from a first-person perspective, the everyday though which the project came about. Devendorf conceptualized, designed, and fabricated the fabric, sensors, and circuitry by hand using processes of stitching, weaving, and felting. These tasks that took place in parallel with many other repetitive tasks and repetitive days spent in close collaboration with a small set of materials and humans. The garment is marked by these manic and memorable experiences as wear upon the material and numeric data collected by the sensors. Specifically, the sensors recorded the forces collected on my body at 21 individual points. The garment, thus, can “play back” these forces as color change in the regions that the force was incurred. The color change is subtle and slowly changing, casting uncertainty and inviting interpretation as to the bodily experience. It lends itself to be read within the frame of a viewer's lifeworld — a Rorschach test more than a data visualization.

We intend to display *Wear* on a hanger suspended from the ceiling or attached to a wall via a pin. Visitors engage the garment by examining the circuitry in the weave and/or by viewing the heat activation through a smart phone equipped with an infrared camera (which we can provide). If no such hanging option is available, the first author will be present at the demo and can wear the garment.

3.1.2 Magnetic Reverberations. This project was produced during the 2023 Experimental Weaving Residency, featuring Elizabeth Meiklejohn, and in collaboration with Irene Posch [12]. By approaching programmable movement as a provocation, this project explored fabrics capable of oscillating between two states. Collectively, we used our knowledge of weaving, electronics, and programming woven drafts to generate a series of samples that fold, flap, and collapse before arriving at a vision for an interactive textile component. The result is an e-textile woven in a single piece that, when removed from the loom, can be cut apart into distinct flaps.

When connected to a controller via a custom interface we built to control its motion, the cloth can perform gestures like rustling, flickering and slow rhythmic opening and closing. The movement of the flap is controlled by electromagnetic coils. When a coil is powered and attracted to a magnet, the flap closes and the cloth appears white. When opened, light bounces from the flap's bright orange interior onto the base cloth, creating a warm neon glow—effectively changing the color of the fabric in a large-scale, structural manner. Through close collaboration and extensive prototyping, we developed weaving strategies in which disparate elements - neutral base, neon flaps, copper coils - are fully integrated into a single-piece fabric on the loom. Designing a woven fabric that not only contains actuators, but lends itself easily to actuation through zones of rigidity and softness, was an equally important part of this process, developing from conversations with engineers and designers in our lab.

While the original piece in the photographs was sadly lost in shipping, we are in the process of recreating it. We envision visitors interacting with the piece by using a tablet controller (which we will provide) or by loading the control software directly on their phone from a QR code.

3.1.3 A Fabric that Remembers. This fabric measures how hard the fabric is being pressed in six different locations in real-time. The wiring for the resistive sensing and voltage dividing is embedded into the fabric via different conductive and resistance yarns. The fabric also readily plugs into a web-connected controller that stores information about how hard, and where the fabric is touched over time. These results can be explored through an online visualization connected to the piece itself, or viewed from their personal device.

This piece is currently held by Accenture Research Labs and we will work with them to loan the piece to us for interactivity. If it is not able to be loaned, we have other woven samples that illustrate the fundamental behavior of measuring press. Visitors will be able to interact with the piece via a tablet we provide or by loading the visualizer on their personal device via QR code.

3.2 Resources

We complement the interactive demos with resources that will allow visitors to more deeply explore experimental weaving in their practice.

3.2.1 AdaCAD: Open Source Parametric Design for Textiles. AdaCAD is a computer-aided design tool that bridges engineering and craft communities to support knowledge sharing and innovation for textile applications [9, 10]. AdaCAD provides a robust and flexible specification for developing and documenting woven textile structures. It currently supports 100 monthly users, has been cited in 78 research publications, and has been distributed through research publications, social media, talks and workshops in both engineering and textile contexts. AdaCAD is the only open-source and/or affordable option for woven structure design that can support complex techniques common in STEM textiles. It is also the only weaving software that leverages parametric design, a form of computational design that uses dataflows of parameterized components to generate objects (in this case, woven structures) that conform to user-defined rules.

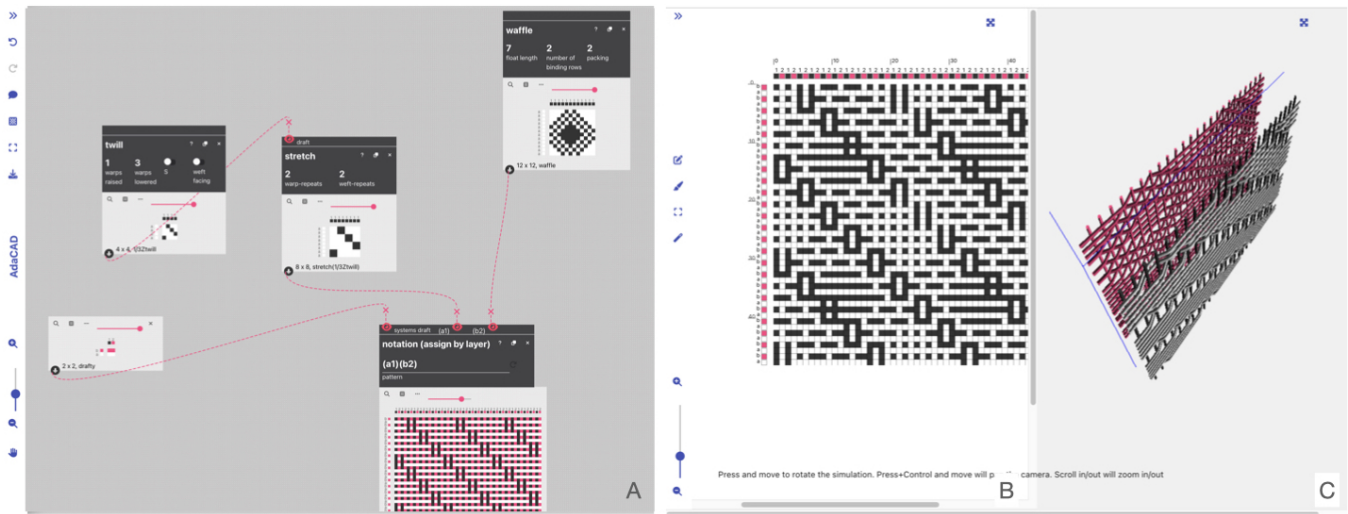


Figure 4: AdaCAD is a computer-aided design tool that bridges engineering and craft communities to support knowledge sharing and innovation for textile applications

In addition to having a monitor and computer running AdaCAD for visitors to play with, we intend for visitors to learn more about the software by first looking at samples that the software was used to make. Each sample will be connected to a QR code. Photographing the QR code loads the design space used to create that sample in the AdaCAD software (which runs in the browser).

3.2.2 Intro to Weave Structure for HCI Workbook. We developed this follow-along workbook with former artist in residency and now PhD student Etta Sandry, with the goal of communicating the foundation of complex 3D weaving to the interactive design community. The workbook was originally published as a Pictorial at DIS 2023 [7] and we will print copies of this text for visitors to take away and complete at their own leisure. The workbook includes descriptions of the weaving process, ways of designing for looms, and hands on paper weaving activities that help people develop an embodied understanding of multi-layer weaving. Participants will be invited to take a workbook or, alternatively, to download and the workbook digitally at a later time.

3.2.3 Experimental Weaving Residency and Talk Series Resources. Most residencies that we have hosted in our research lab have been commemorated with a printed catalog that describes the research explored during the residency and documents the many samples, ideas, and innovations that emerged through the collaboration. We have created two catalogs to date: one showcasing the design of woven dry electrodes with artist Sandra Wirtanen and the other documenting the creation of Magnetic Reverberations with Elizabeth Meiklejohn. Visitors to the demo will be able to view the physical catalogs (which they can also view online). Additionally, our research labs hosts a series of talks that feature experimental weavers describing their process and projects. We'll bring attention to these talks by having a tablet screen dedicated to replaying our playlist of recorded talks (all of which are accessible via YouTube for later viewing).

4 Conclusion

We look to bring more awareness and resources to the CHI community around the relevance of weaving to interactivity. We do this by showcasing the programs and products of our research, specifically those that have involved collaborations with "experimental weavers." This will be the first time this collection of resources has been compiled outside of our research lab. As members of the CHI community, we are passionate to share our processes and findings with this community in a format that can provoke interest and further inquiry.

Acknowledgments

The work we will show has been made possible by collaborations with a large number of people. We'd like to specifically acknowledge the key role of our experimental weavers in residence, including Elizabeth Meiklejohn (for contributions to AdaCAD, samples, and leading the Magnetic Reverberations project), Kathryn Walters (for contributions to AdaCAD), Etta Sandry (for contributions to AdaCAD as well as the weave structure workbook), and Sandra Wirtanen (for contributions to AdaCAD and samples for display). Additionally, we would like to acknowledge the contributions of key collaborators, especially Irene Posch (for collaborating on Magnetic Reverberations and teaching resources), Marianne Fairbanks (for contributions to AdaCAD) and Emma Goodwill (for contributions to AdaCAD). This work we will showcase has been supported by National Science Foundation grants #2405297, #2346150, and #2413631, the Center for Craft, and Accenture Research Labs.

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